S. Shayan Daneshvar

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Professional Summary

LLM research engineer specializing in post-training for agentic coding: verifiable SWE environments, rejection-sampled distillation, SFT, and multi-node inference and evaluation for dense/MoE models, including the open-source Pangu 92B. First-author papers in ACM TOSEM (Q1) and EASE (CORE A), plus strong production software engineering experience.

Technical Skills

LLM Post-Training: SFT, Rejection Sampling, Knowledge Distillation, DPO, RLVR, LoRA
Distributed Training & Inference: Megatron-LM, LLaMA-Factory, DeepSpeed ZeRO, 3D Parallelism, vLLM, Ray
AI Agents & Evaluation: LangGraph, LangChain, MCP, Agentic RAG, Harbor, Weave, G-Eval, Claude Code, Codex
Machine Learning & Deep Learning: PyTorch, Hugging Face Transformers, OpenCV, WandB, Computer Vision, NLP, Domain Adaptation, Data Augmentation, Diffusion Models, Synthetic Data Generation, Contrastive Learning
Programming Languages: Python, Java, C/C++, SQL, Bash
Software Engineering: Spring Boot, Microservices, Docker, Kubernetes, Kafka, NoSQL, Git, CI/CD, AWS, Scrum/Agile

Professional Experience

LLM Research Engineer Nov 2025 – present
Huawei Technologies Canada *Kingston, Canada*

- Proposed and built a task augmentation pipeline that grew rejection-sampled feature-implementation SFT data by **50%**, lifting Qwen3-14B by **4.3 pp on SWE-bench Verified** and **7.1 pp on FeatBench**
- Architected a multi-agent SWE environment builder with **self-evolving executable procedural memory**, building 10K+ environments in 10 languages and beating SWE-Gen with **2.2× yield**, 3.8× throughput, 68% fewer tokens
- Designed multi-stage guidance for **agent evaluation** to reduce false failures from hidden API mismatches, yielding **fairer benchmarking** and graded training signals for PPO/DPO
- Built a trajectory quality analyzer that classifies agent tool calls by phase to flag loops, brute-forcing, and skipped verification, marking **12%** of issue-resolution trajectories for redistillation
- Ran continual **supervised fine-tuning** of **Pangu 92B MoE** on 1B+ tokens of my distilled trajectories plus a subsample of its original code data, across multi-node distributed GPU/NPU clusters

Machine Learning Engineer Apr 2025 – Sep 2025
MacDon Industries (Contract via Zoomi Technologies) *Winnipeg, Canada*

- Led end-to-end delivery of a DAFormer-based unsupervised domain adaptation model with a custom ClassMix, beating the supervised baseline by **3 pp** in Dice score using only unlabelled real data, in 75% of the planned time
- Customized a diffusion method with **ControlNet** to generate labelled synthetic images, cutting real labelled data needs by **70%** while maintaining semantic segmentation quality

Machine Learning Engineer Intern Nov 2024 – Mar 2025
MacDon Industries (Mitacs Accelerate) *Winnipeg, Canada*

- Integrated and trained SegFormer for a **4 pp Dice** gain, and diagnosed the real-vs-synthetic domain gap with **UMAP** on **CLIP** and **VGG embeddings**, steering the project to domain adaptation

Graduate Research Assistant Jan 2023 – Mar 2025
University of Manitoba *Winnipeg, Canada*

- First-authored VulScribeR (**TOSEM**), the first use of **LLMs** for vulnerability data augmentation, using **RAG** to lift detector F1 by ∼**28%** over SOTA generators, at US$1.88 per 1K samples
- First-authored the first systematic study of mixup-based augmentation for vulnerability detection (**EASE**)

Senior Software Engineer May 2022 – Dec 2022
Mahsan *Tehran, Iran*

- Guided a **team of 4** in architecting a distributed data-analytics platform combining **microkernel** and **microservices** architectures with **CQRS**, **Saga**, and **Event Sourcing**
- Built **distributed tracing** and **monitoring** for all microservices with Spring Cloud Sleuth, Zipkin, Prometheus, Grafana, Elasticsearch, Kibana, and Logstash
- Conducted 50+ code reviews enforcing coding standards, mentored 2 junior developers, and taught Spring Cloud and microservices to **7+ developers** over 30+ hours

Backend Software Engineer Nov 2021 – May 2022
Tosan Soha *Tehran, Iran*

- Developed hybrid Java/Kotlin e-wallet and trading platforms on **Kubernetes** serving **2K+ concurrent users**, with banking APIs and event-driven Kafka workflows
- Drove the migration from a tangled JWT/session login to **Spring Security OAuth** for third-party login, proposing it and implementing most of the OAuth flow and cross-service refactoring

Java Backend Developer May 2021 – Nov 2021
Pinket *Tehran, Iran*

- Developed event-driven chat, notification, and shopping services with Spring, Kafka, PostgreSQL, and MongoDB
- Maintained the chat service and added integration and unit tests to CI/CD, raising code coverage by **7%**

Java Instructor and Mentor (Part-time) Jan 2020 – May 2021
Mapsa HR and Training *Tehran, Iran*

- Taught core and advanced Java, including OOP, SQL, data structures, concurrency, and algorithm design
- Delivered hands-on training and mentorship in Spring Boot, Spring MVC, JPA, Docker, and REST APIs

Education

University of Manitoba Winnipeg, Canada
Master of Science in Computer Science, Deep Learning for Software Engineering, 4.5/4.5 GPA

- **Thesis**: Exploring Representation-level Augmentation and RAG-based Vulnerability Augmentation with LLMs for Vulnerability Detection
- **Supervisor**: Dr. Shaowei Wang

K. N. Toosi University of Technology Tehran, Iran
Bachelor of Science in Computer Engineering, 4/4 GPA, ranked 1st of 76

- **Thesis**: Reflection Removal of In-vehicle Images with UNets
- **Advisor**: Dr. Behrooz Nasihatkon

Research Publications

VulScribeR: Exploring RAG-based Vulnerability Augmentation with LLMs 2026
***S. Daneshvar**, Y. Nong, X. Yang, S. Wang, H. Cai – ACM TOSEM (Q1, top software engineering journal)*

A Study on Mixup-Inspired Augmentation Methods for Software Vulnerability Detection 2025
***S. Daneshvar**, D. Tan, S. Wang, C. Leung – EASE 2025 (CORE A)*

GUI Element Detection Using SOTA YOLO Deep Learning Models 2024
***S. Daneshvar**, S. Wang – arXiv preprint*

The application of barcode readable assay and linear regression RGB analysis... 2024
*N. Rezaei, **S. Daneshvar**, B. Nasihatkon, S. Seidi, M. Rezazadeh – Microchemical Journal (Q1)*

Honors & Service

Service: Reviewer, ACM TOSEM (2026–present) • ICSE 2026 Shadow Program Committee
Awards: UManitoba Graduate Fellowship • Research Completion Award • International Graduate Entrance Scholarship
Distinctions: Ranked 1,427th of 144,437 (top 1%) in Iran's national university entrance exam • Dean's List 2019

Projects

Reflection Removal with Mamba/S6 State-Space Model | *Python, Docker, OpenCV, PyTorch* 2024

- Built ImageMamba, an adapter that runs the original Mamba state-space module on image feature maps, to replace attention in the SOTA DSRNet, beating it on PSNR (**+0.14 dB**) and SSIM

Chibaladi.com | *Spring Cloud, PostgreSQL, MongoDB, Spring Data, Nginx* 2021

- Independently designed, built, and deployed a microservices e-learning platform with video courses and quizzes

Languages

English (Fluent, C1 – CELPIP 11/12) • Persian (Native)